RIG-ETALE-NESS OF A CONTINUOUS HOMOMORPHISM IS INDEPENDENT OF THE IDEAL OF DEFINITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces the two ideals of definition to each other by commensurability and invokes the independence of the completed naive cotangent complex

Editorial note. The label lemma-rig-etale-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the finite-type conditions of Lemma lemma-finite-type in the category WAdm^Noeth
- the definition of a rig-etale homomorphism of adic rings (definition-rig-etale-homomorphism) via the completed naive cotangent complex
- independence of the completed naive cotangent complex from the choice of ideal of definition (remark-NL-well-defined-topological)

Source reference. Original reference 0GAY: sources/stacks-restricted/pinned-original-source.tex, source lines 1794–1794. Prerequisite pointer invoked at proof line(s) 4768; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0ANA: sources/stacks-chow/tags.txt, source lines 13433–13433. Prerequisite pointer invoked at proof line(s) 4751; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 04U5: sources/stacks-restricted/pinned-original-source.tex, source lines 2439–2439. Prerequisite pointer invoked at proof line(s) 4755; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GBX: sources/stacks-restricted/pinned-original-source.tex, source lines 2487–2487. Prerequisite pointer invoked at proof line(s) 4772; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $A \to B$ be a morphism in WAdm^{Noeth} (Formal Spaces, Section 0ANA). The following are equivalent:*

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(a) *$A \to B$ satisfies the equivalent conditions of Lemma 04U5 and there exists an ideal of definition $I \subset B$ such that B is rig-étale over (A, I), and*
(b) *$A \to B$ satisfies the equivalent conditions of Lemma 04U5 and for all ideals of definition $I \subset A$ the algebra B is rig-étale over (A, I).*

Proof. Let I and I' be ideals of definitions of A. Then there exists an integer $c \geq 0$ such that $I^c \subset I'$ and $(I')^c \subset I$. Hence B is rig-étale over (A, I) if and only if B is rig-étale over (A, I'). This follows from Definition 0GAY, the inclusions $I^c \subset I'$ and $(I')^c \subset I$, and the fact that the naive cotangent complex $NL^\wedge_{B/A}$ is independent of the choice of ideal of definition of A by Remark 0GBX. □